

Chapter 5

From Macro to Micro Architecture

5.1 Overview

The circulation works at two scales. In the macrocirculation, the large arteries, each heartbeat sends a pressure wave from the heart to the periphery, and the timing of that wave carries information about blood pressure. In the microcirculation, small vessels match local supply to local demand. In the brain this matching is tuned continuously. Astrocytes help raise flow where neurons are active, collateral vessels take over when a route is lost, and vessels that carry little flow are gradually pruned. This chapter draws on both scales. The physics of the large arteries defines what a correct blood pressure model must compute, and the regulation of the microcirculation suggests how a model should allocate its own computation.

Blood pressure is the motivating measurement. Wearable devices now stage sleep, estimate heart rate from light passed through the skin, and remove motion noise as it happens, all continuously on a device the size of a watch face. Blood pressure, one of the most basic hemodynamic measurements, is still taken much as it was around 1900. A cuff is inflated around the upper arm until it stops the flow of blood, then slowly released while the device listens for the sounds, or senses the pressure oscillations, of blood forcing its way back through the compressed artery. The result is a single reading of a quantity that changes with every heartbeat. The continuous reference is a catheter with a pressure sensor placed inside an artery, used only in intensive care and during surgery. Neither allows continuous monitoring in daily life.

Pulse transit time offers a way around both. Timing the pressure wave between two points on the body gives its speed, and the wave travels faster when pressure is higher. The artery, however, is not a passive pipe. When the left ventricle ejects blood into the aorta, the blood itself moves at about 1 m/s at peak systole, while the stretch of the arterial wall travels along the arterial tree at 4 to 15 m/s, depending on how stiff the arteries are. Transit time measures this wave in the wall, and the stiffness of the wall depends on pressure in a way that differs between people (Section 5.2). A cuffless estimator can therefore match reference pressure on a benchmark while computing nothing related to pulse propagation, and accuracy alone cannot reveal this.

Chapter 3 constrained what a model sees and how it decides, restricting the inputs to hemodynamic and motion sensors and reducing a gradient-boosted model to a cascade a person can read. Chapter 4 used the macrocirculation to constrain the macro architecture of a model, at the level of cross-attention between measurement sites. This chapter moves inside the model. Project 3A (Section 5.3) uses the physics of the large arteries to verify a model: a mechanistic

audit tests whether an estimator’s prediction depends on transit time in the way the physics requires, with preliminary results on a synthetic testbed. Project 3B (Section 5.4) uses the regulation of the microcirculation to build the model: a rule we call hemodynamic attenuation closes attention heads during training, the way unneeded vessels regress, until only the circuit the task needs remains, bringing a deep model closer to the readability of the Chapter 3 cascade. The two meet in a joint experiment. A blood pressure model trained with hemodynamic attenuation keeps only a few heads, and the audit of 3A can then ask which of them, if any, carries transit time (Section 5.4.9).

5.2 Background

5.2.1 Transit time and pressure

The speed of a pressure wave in an elastic tube is given by the Moens–Korteweg relation, $PWV = \sqrt{Eh/(\rho d)}$, where E is the elastic modulus of the wall, h its thickness, d the vessel diameter and ρ the density of blood. The wall stiffens as pressure rises, which is commonly modelled as $E = E_0 e^{\zeta P}$ [1]. Combining the two gives

$$PWV = \sqrt{\frac{E_0 h}{\rho d}} e^{\zeta P/2}, \quad P = \frac{2}{\zeta} \ln PWV - \frac{1}{\zeta} \ln \frac{E_0 h}{\rho d}. \quad (5.1)$$

Pressure is a logarithmic function of wave speed with constants that differ between people, which is why transit-time methods need calibration [2].

5.2.2 Arrival time is not transit time

Most cuffless systems measure pulse arrival time (PAT), from the R-peak of the ECG to the foot of the pulse in a peripheral PPG. This interval also contains the pre-ejection period, the delay between the electrical activation of the heart and the opening of the aortic valve:

$$PAT = PEP + PTT. \quad (5.2)$$

PEP changes with contractility and sympathetic tone independently of arterial pressure, so any model trained on PAT inherits this confound. For a site at path length d_i from the heart, $PAT_i = PEP + d_i/PWV$, so the difference between two sites cancels PEP. Deep networks trained on raw ECG and PPG inherit the confound without showing it, and may also use regression toward the population mean, demographic priors, or waveform shape unrelated to propagation.

5.2.3 Accuracy cannot certify mechanism

Two estimators with the same error on a dataset can compute different functions of their input. One routes its prediction through transit time. The other routes it through confounds that happen to vary with pressure in that dataset. The two diverge only under distribution shift, which is the condition of deployment. Verifying a model therefore requires intervening on it, not only evaluating its outputs.

5.3 Project 3A: A Mechanistic Audit for Blood Pressure Estimators

The audit asks whether an estimator’s prediction depends on transit time in the direction and magnitude that Equation (5.1) requires. It has three tiers of increasing strength.

1. Linear probes test whether PTT can be read linearly from the model’s internal activations.
2. Nonlinear probes test whether it can be read at all, separating absence from nonlinear encoding.
3. Causal tests intervene on the model. PTT-direction ablation removes the decoded direction and measures the change in prediction. Donor-swap activation patching transplants activations from an input with a different transit time and checks that the prediction moves by the amount the physics predicts. A roll test delays the distal PPG, which lengthens PTT, so the predicted pressure must fall. Each effect must have the correct sign and a physically plausible size.

A model that is accurate but fails every tier is classified as not encoding transit time. At run time such a model abstains rather than reporting a pressure it cannot justify.

Synthetic testbed. Real data cannot validate an audit, because the true mechanism of a trained network is unknown. We therefore built a testbed in which it is known. Part 1 is a scalar task solved by three small networks with different, known mechanisms. Part 2 is a time-domain tube-load waveform simulator in which foot-to-foot PTT was verified to be unchanged by the load parameter Γ .

Preliminary result. Across the Part 2 sweep, the donor-swap effect increased steadily with α , the degree to which the model relies on transit time, while accuracy stayed flat. Accuracy did not change and the audit did, which is direct support for the argument that accuracy cannot certify mechanism.

Remaining work. Part 3 applies the audit to published architectures trained on real ECG and PPG. Whether most such models fail the audit is the hypothesis of that experiment, not yet a result. We also piloted a camera-based measurement of transit time between the face and the hand. The remote PPG signals were too noisy to give reliable transit times, and camera-based systems of this kind have already been reported [3], so this route is not pursued.

5.4 Project 3B: Hemodynamic Attenuation of the Micro Architecture

An audit is only practical when the computation it examines is small. In the microcirculation, supply is not fixed in advance. Vessels are kept where they are needed and pruned where they are not. Project 3B applies the same idea to attention heads. In Chapter 4, a trained transformer carried its causal computation in two of forty heads, but this was found only after training, by ablating each head in turn, and the number of heads to keep had to be chosen by hand. Project 3B asks whether a model can find the number of heads it needs during training, without

being given it. We introduce a rule for this, hemodynamic attenuation, and test it on tasks whose minimal number of heads is known.

5.4.1 Motivation from cerebral blood supply

Four properties of cerebral blood supply motivate the rule:

1. Supply follows need. Local blood flow rises with neural activity, signalled in part by astrocytes [5]. Each head is given a valve $g_h \in [0, 1]$.
2. A vessel is dispensable when collateral vessels can supply its tissue [6]. A head is valued by what is lost when it closes and the other heads compensate, which we call its collateral value.
3. Maintaining a vessel has a metabolic cost [7]. Each open head has a fixed price, and there is no target number of heads.
4. Unneeded vessels regress gradually [8, 9], and collaterals can enlarge when a route is blocked [6]. Closing heads fade out over 100 training steps and can be reopened.

The biology motivates the design, and no claim is made about the brain.

5.4.2 Astrocytes and attention

Softmax attention turns a query’s scores $s_1, \dots, s_N$ into weights

$$a_i = \frac{e^{s_i}}{\sum_j e^{s_j}}, \quad (5.3)$$

a numerator that differs for each memory slot and a denominator that is one number shared by all of them. Kozachkov et al. [4] showed that a network of neurons and one astrocyte can approximately compute self-attention. The neurons store the numerators in Hebbian synapses. The astrocyte computes the denominator. It pools activity across all the synapses it wraps and scales every one of them by the same gain, $1/p_t$. For token t the output is approximately (their Eq. 11)

$$\ell_t \approx \frac{1}{p_t} \sum_i e^{k_i^\top q_t} v_i + x_t, \quad p_t = \sum_j e^{k_j^\top q_t}, \quad (5.4)$$

which is softmax attention plus a residual connection. The point used here is that the astrocyte acts as a single gain on a whole block of synapses.

5.4.3 Head pruning

Michel et al. [13] remove heads by a gradient score, and Voita et al. [14] learn head gates with an L_0 penalty. Optimal Brain Damage [10] scores a parameter by the loss increase when it is removed, and Optimal Brain Surgeon [11] by the increase after the remaining parameters are re-fitted. ZipLM [12] applies the latter to whole heads of trained language models. All of these prune after training, to a size chosen by the user.

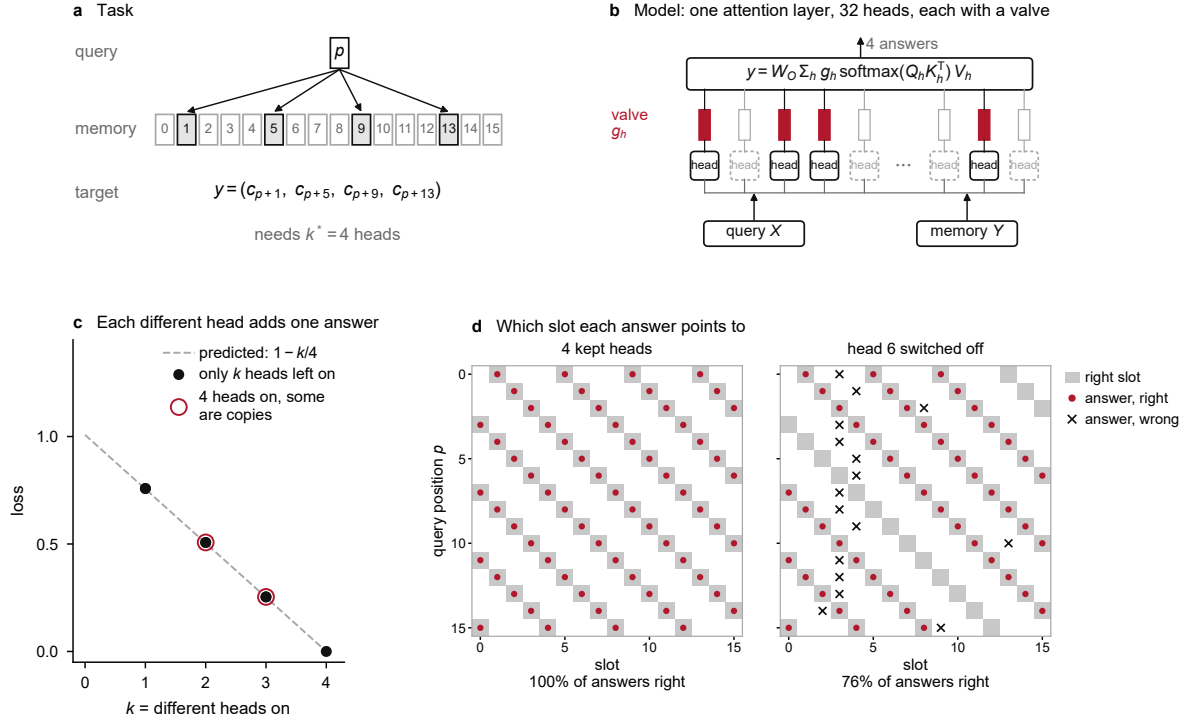


Figure 5.1: Task and model. **a**, A query carries a position p , and the target is the items stored 1, 5, 9, 13 slots ahead, so 4 heads are needed. **b**, One attention layer with 32 heads, each with a valve g_h . Filled valves are open. **c**, Loss with k different heads on a trained 4-head model, against the prediction $1 - k/4$. **d**, The slot each answer points to. All are correct with the 4 kept heads, and 76% with head 6 switched off.

5.4.4 Task

The memory has $N = 16$ slots, each holding a random item c_j . A query specifies a position p , and the target is the items at R fixed offsets (Figure 5.1a):

$$\text{target}(p) = (c_{p+\delta_1}, \dots, c_{p+\delta_R}), \quad \text{slots modulo } N. \quad (5.5)$$

Items are redrawn for every example, so the model must learn where to attend.

A head returns, for each query, one weighted average of the items, which is one linear equation in the R unknown items. Recovering R items needs R independent equations, so the task needs $k^* = R$ heads, and with $k < R$ heads the lowest achievable loss is

$$\text{loss}(k) = \left(1 - \frac{k}{R}\right) L_{\text{triv}}, \quad (5.6)$$

where L_{triv} is the loss of predicting the mean. Duplicate heads give the same equation and count once. Measured losses agree with Equation (5.6) to within 1% (Figure 5.1c). The task counts as solved when the loss is below $0.02 L_{\text{triv}}$. Since L_{triv} is the variance of the targets, this is the same as $R^2 \geq 0.98$, the model explaining 98% of the variance in the answers. Results are unchanged for any threshold from 0.01 to 0.1.

5.4.5 Model

The model is one cross-attention layer with $H = 32$ heads and no MLP. Matrices have one column per token, as in [4], with X holding the queries and Y the memory. Head h computes the usual attention,

$$O_h = V_h \text{softmax}\left(\frac{K_h^\top Q_h}{\sqrt{d_k}}\right), \quad Q_h = W_Q^h X, \quad K_h = W_K^h Y, \quad V_h = W_V^h Y, \quad (5.7)$$

with the softmax taken down each column, so each query’s weights over the memory add up to one. Each head’s output is multiplied by its valve g_h before the heads are added together:

$$\hat{T} = \sum_{h=1}^H g_h W_O^h O_h. \quad (5.8)$$

A head with $g_h = 0$ contributes nothing and receives no gradient, so a model with k open valves is exactly a k -head model. The valve plays the same algebraic role as the astrocyte gain, one number that multiplies a whole block of synapses, here the output weights W_O^h of one head. The two differ in what sets them and how fast they change. The astrocyte gain follows pooled input activity and changes with every token to normalise attention, while the valve follows the head’s collateral value and changes slowly over training to select heads. The correspondence is algebraic only.

5.4.6 Hemodynamic attenuation

All valves start open. After a warm-up of 10% of training, every 25 steps the rule (1) computes the collateral value of every open head, (2) closes the lowest-valued head if its value is below the price $0.03 L_{\text{triv}}$, and (3) reopens the highest-valued closed head if its value exceeds twice the price. A closing valve decreases by 1/100 per step.

The collateral value is the increase in the best-fit loss when head h is removed and the other open heads re-fit their output weights:

$$v_h = \min_W L(\text{without } h) - \min_W L(\text{all open heads}). \quad (5.9)$$

This is the Optimal Brain Surgeon score [11], applied to whole heads as in ZipLM [12], and it is computed in closed form from one matrix inverse. The score is not new. The contribution is its use during training, with a price, a gradual fade, and reopening. The re-fit is what makes a copy worthless. If two heads output the same signal with weights (0.5, 0.5), removing one without re-fitting halves the output, so both look important. With re-fitting the other head takes weight 1, the loss does not change, and the copy is worth nothing.

5.4.7 Baselines and evaluation

All methods share the model, task, training and runs. The published methods are run inside our training: ZipLM-style pruning [12] (the same score applied once, then fine-tuning), Michel et al. [13], and Voita et al. [14]. Three controls each remove one part of the rule: no fade, no re-fit (the Optimal Brain Damage score [10]), and no score (random choice). We report exact count (open heads over the last 40% of training equal k^*) and never broke (after first falling below the solved threshold, the loss never exceeds it), on $k^* = 2, 4, 8$ with 10 seeds each, together with runs with duplicated heads, runs with random head failure, and a two-layer induction task [15].

5.4.8 Preliminary results

Count and stability. Hemodynamic attenuation kept exactly k^* heads in 29 of 30 runs and never broke the model in 30 of 30. The count does not depend on the price. Every price from 0.01 to 0.1 of L_{triv} gave the exact count (9 runs each), because a needed head is worth at least L_{triv}/k^* and a redundant head close to zero, so any price between the two gives the same answer. No other method achieved both (Figure 5.2c). The method of Michel et al. stops when the loss crosses the solved threshold, so it breaks the model by design. Told the true number of heads instead, it stopped at that number but still broke the model in 28 of 30 runs. Without the fade the model broke in every run. Without the re-fit it kept too many heads. With random choice it broke in 24 of 30 runs.

Duplicated heads. With every head starting as a copy of another, the rule kept 4 heads and never both members of a pair. Without the re-fit, 8 to 15 heads and 4 to 7 pairs were kept (Figure 5.2d).

Backup heads. When heads fail at random, the rule should keep extra heads in proportion to how likely they are to be needed. A formula written before the experiments predicted the number kept exactly in 21 of 24 runs and to within one head in the rest (Figure 5.2e).

Induction. In a two-layer attention-only model with $8 + 8$ heads, the minimal induction circuit is one previous-token head in layer 1 and one induction head in layer 2 [15]. This is a classification task, so solved uses the classification form of R^2 , McFadden’s pseudo- R^2 , one minus the model’s cross-entropy divided by that of random guessing among the 64 tokens [16]. The task counts as solved when it is at least 0.98, which corresponds to about 99% of the predictable tokens correct. Our runs confirm this minimum (Figure 5.3c). A model with $1 + 1$ heads trained from scratch solved the task in 10 of 10 runs, while a single layer failed in every run, whether it had 1 head or 8. The circuit needs depth, not more heads. A model with $2 + 1$ heads also solved the task in 10 of 10 runs, with no lower loss than $1 + 1$ (median 0.0037 against 0.0033). Hemodynamic attenuation kept $2 + 1$ heads in 10 of 10 runs and never broke the model. Layer 2 reached the needed single head, while layer 1 kept one head more than needed (Figure 5.3b). In one run examined in detail, the two layer-1 heads had each learned a weak version of the previous-token job, placing 28% of their attention on the previous token. Together they gave 99.9% correct predictions on the repeat, either one alone 86 to 88%, and neither 6%. Closing either head would have cost far more than the price, so the rule kept both rather than break the model while the remaining head sharpened. Switching off the kept layer-2 head reduced next-token accuracy on the repeat to zero (Figure 5.3a).

Cost. The final loss was 2 to 7 times the dense model’s, well below the threshold, using 0.31 to 0.45 of its head-steps. The saving is nominal, since closed heads are still computed.

5.4.9 Joint experiment: attenuate, then audit

The two projects meet in one experiment. A transformer is trained to estimate blood pressure on the synthetic tube-load testbed of Project 3A, where the role of transit time is known, with hemodynamic attenuation closing heads during training. The audit of Project 3A is then run on each of the few heads that remain. If the model computes pressure from transit time, the audit

should find it in a small number of named heads, and switching those heads off should move the predicted pressure in the direction the physics requires. If no surviving head carries transit time, the model is accurate for another reason, and the audit has caught it with far fewer heads to check.

5.4.10 Proposed work

The proposed work follows the timeline in Figure 5.4. It begins with the joint experiment of Section 5.4.9, which trains a blood pressure model with hemodynamic attenuation and audits the heads that remain.

The rule currently values a head by re-fitting only the final read-out. Between layers, one layer’s heads change what the next layer attends to, and a re-fit at the output cannot repair that. This is why layer 1 of the induction model keeps one head more than needed. The proposed extension values a head at its own layer, by asking the other heads of that layer to rebuild what the next layer reads, the same layer-wise approach used to prune large language models [12]. It will be developed on small language models trained from scratch, where accuracy can be measured against the number of open heads, and checked on the induction task, where it should reach the needed $1 + 1$ heads.

The rule will then be applied to the models of Chapter 4, asking whether it keeps the two causal heads found in qSA-Cross, and whether the same heads are kept across all sixteen leave-one-subject-out folds of the Cross-Site Transformer on WildPPG. Finally, it will be applied while fine-tuning GPT-2 small on indirect object identification, a task whose circuit in that model is known [17], asking whether the heads the rule keeps overlap with that circuit.

5.5 Timeline

5.6 Risks and Fallbacks

The audit is inconclusive on real data. The audit’s contribution does not depend on new human data. If published models cannot be retrained or their activations are unavailable, the audit is reported on the synthetic testbed and on models trained here on public multi-site recordings.

The rule does not reach the minimum in deeper models. This is already visible in the induction task, where layer 1 keeps one extra head. If the layer-wise value does not close the gap, the result is still a small circuit found during training, and the gap is reported with the number of extra heads per layer.

The rule does not scale. The collateral value needs one matrix inverse per layer. If this is too costly for large models, it will be computed on a subset of layers or with the approximations used in large-model pruning, and the accuracy of the approximation will be measured on the small tasks where the exact value is known.

The kept heads differ from those found in Chapter 4. This would be a finding about how stable circuits are across training methods, and will be reported as such.

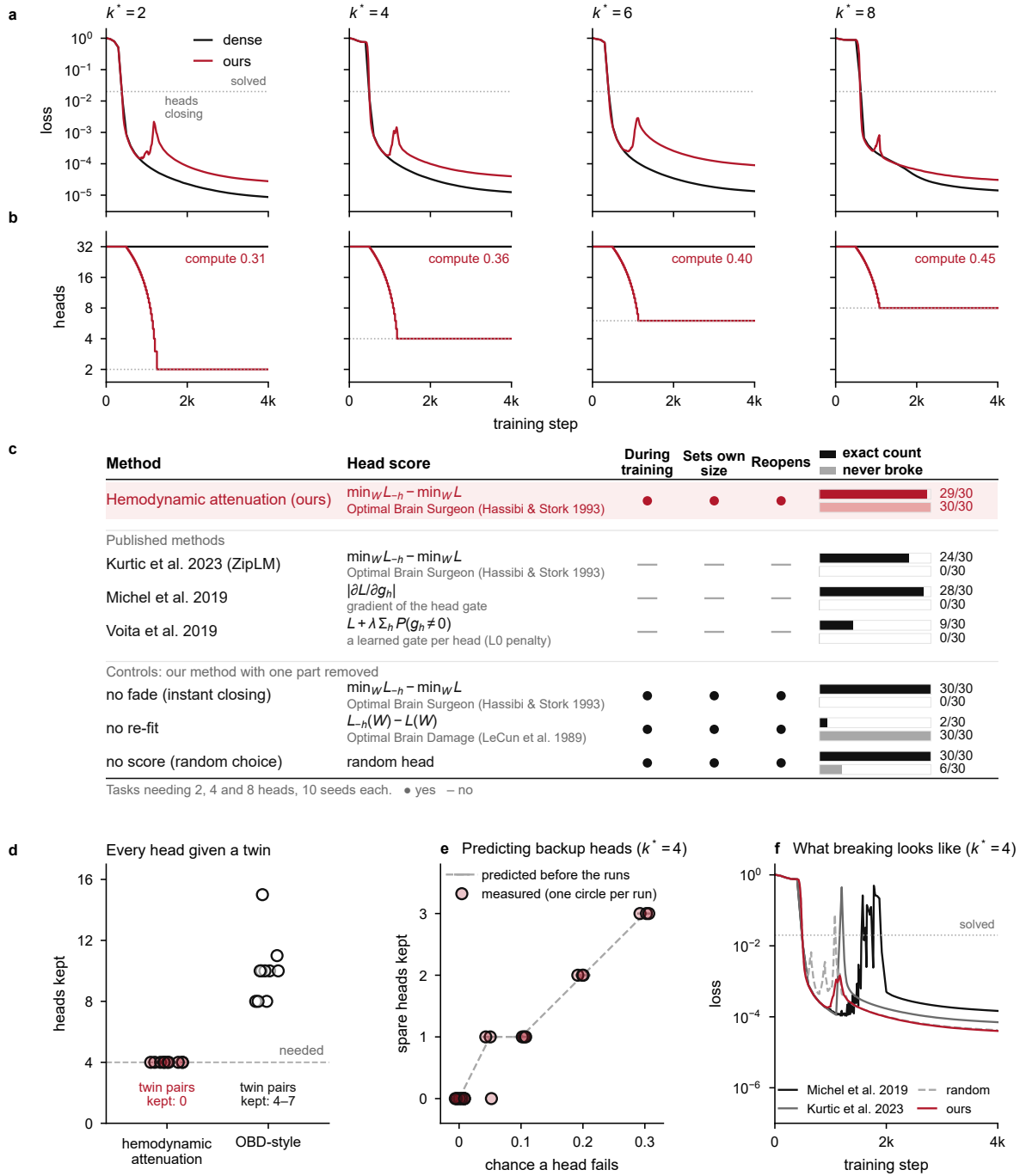


Figure 5.2: Hemodynamic attenuation during training. **a**, Loss of the dense model (black) and with hemodynamic attenuation (red). **b**, Open heads, with k^* dotted. **c**, All methods on the same 30 runs: whether each prunes during training as published, sets its own number of heads without a target, and can reopen heads, with how often it kept exactly k^* heads (dark bar) and never broke the model (light bar). **d**, Runs in which every head starts with an identical copy. **e**, Spare heads kept under random failure, against the prediction. **f**, One run per method. Only hemodynamic attenuation stays below the solved threshold.

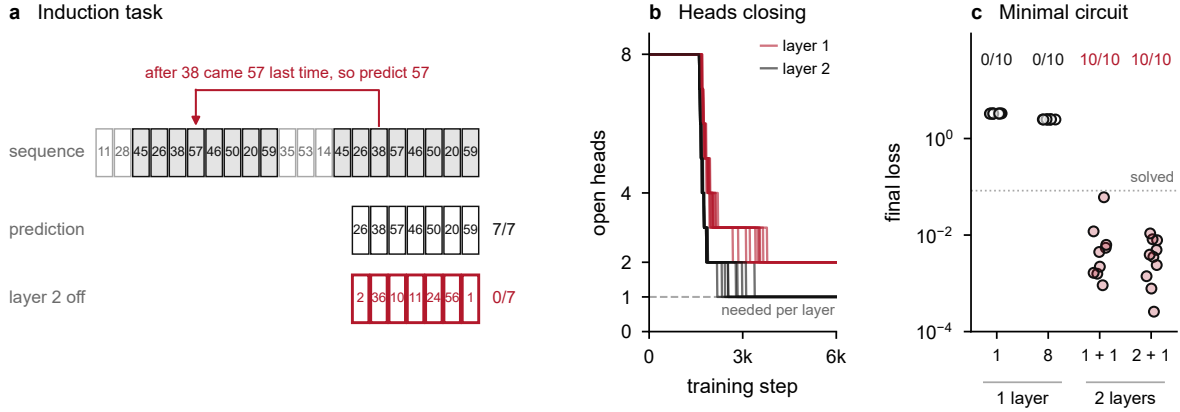


Figure 5.3: Two-layer induction. **a**, An example sequence and the model’s predictions on the repeat, with all kept heads (7/7 correct) and with the layer-2 head switched off (0/7). **b**, Open heads in each layer during training, 10 runs. Layer 2 reaches the needed one head, and layer 1 stops at two. **c**, Final loss of models trained from scratch with only the heads shown per layer, 10 runs each. One layer fails with 1 or 8 heads. Two layers solve the task with one head each, and a second layer-1 head gives no lower loss.

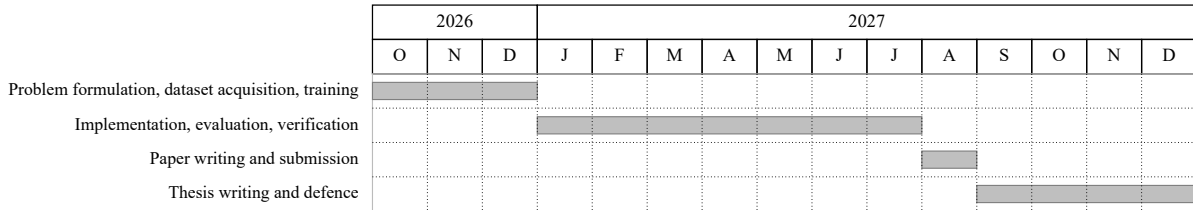


Figure 5.4: Timeline from the proposal to the defence.

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